

HHP – HULDRA HALEPRODUKSJON

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Document title: SYSTEM CONTROL DIAGRAM - FUEL GAS SYSTEM SEQUENCE LOGIC		
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Sequence 1, Start Fuel Gas System Line A.

Phase	Step	Actions	Outputs to	Next step	Transition Condition	Jump condition	Jump to step	Remarks
Start permissive	0	1. No S/D active 2. Liquid in Scrubber 3. Inlet isolation valve open 4. Line A heater closed 5. Line B heater closed 6. System pressurized	NA	2	1. not UZS-4513.Y 2. not 45-LIC4826.WL 3. 45-HS4820.BCH 4. 45-XY4848.BCL 5. 45-XY4850.BCL 6. 45-PIC4831.YX> FUEL_GAS_SP	ELSE	1	Start order to be confirmed by operator. FUEL_GAS_SP can be adjusted by operator, default value 13,7 barg.
System not available	1	1. Warning on OS	1. 45-XA 9299	END	NA	NA	NA	
Set System in auto	2	1. Set system in auto	set auto pin high on: 1. 45-XY4848 2. 45-XY4850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-LIC4827 8. 45-HS4822 9. 45-LIC4826 10. 45-HS4821 11. 45-LIC4828 12. 45-HS4823 11. 45-PIC4831 12. 45-HS4820 14. 45-TDIC4848 15. 45-TDIC4850 16. 45-XV4804 17. 45-XL4805	3	NA	NA	NA	
Check Line B spill-off	3		1. 45-HS4830.XL	4	1. 45-HS4830.BCL	NA		
Check Line A block valves	4	1. Open to Metering 2-4. Open to liquid outlet. 5-6. Check that level control is in auto	1. 45-HS4824.XH 2. 45-HS4822.XH 3. 45-HS4821.XH 4. 45-HS4823.XH 5. 45-XV4804.XH 6. 45-XV4805.XH	5	1. 45-HS4824.BCH 2. 45-HS4822.BCH 3. 45-HS4821.BCH 5. 45-LIC4826.BA 6. 45-LIC4827.BA 7. 45-XV4804.BCH 8. 45-XV4805.BCH	N/A	N/A	Normally this sequence is run after the stop or pressurization sequences. In that case these valves are already opened.
Check Line B block valves	5	1. Close to Metering	1. 45-HS4825.XL	6	1. 45-HS4825.BCL	NA	N/A	
Open Line A spill-off	6		1. 45-HS4829.XH	7	1. 45-HS4829.BCH			
Start Heater A	7		1. 45-XY4848.XP1H	8	1. 45-XY4848.BCH	NA	NA	
Wait for stabilization	8			9	1. 45-TDIC4848.YX=.YR			Within 5%
Wait for stabilization	9			10	1. 45-TDIC4848.YX=.YR 2. timer=60s	NA	8	Timer value to be adjusted

Wait for Compressor purge and inlet valves to open	10	1.Indicate Fuel gas available to OS	1. 45-XA9201.X	11	1. 27-HS4806.BCH 2. 27-ZSH4005.BCH		Override 45-HS4829.BCL to indicate that Line A is up and fuel gas available. The program should anticipate the inclusion of the following measures to avoid an unacceptable fluctuation in gas temperature when turbine starts. 1. Temporary change in PID parameters (45-TDIC4848 and 45-TDIC4850). 2. Temporary change in set points.
Close Line A spill-off	11	1. Close Line A spill-off line	1. 45-HS4829.XL	12	1. 45-HS4829.BCL	N/A	27-HS4006.BCH indicates flow either to flare or to turbine
Reset Forced auto	12		Release Auto pin on 1. 45-XY4848. 2. 45-XY4850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-LIC4827 8. 45-HS4822 9. 45-LIC4826 10.45-HS4821 11.45-LIC4828 12.45-HS4823 11.45-PIC4831 12.45-HS4820 13.45-HS4806 14.45-TDIC4848 15.45-TDIC4850	END			
END	END	NA	NA	NA	NA	NA	NA

Sequence 2, Start Fuel Gas System Line B.

Phase	Step	Actions	Outputs to	Next step	Transition Condition	Jump condition	Jump to step	Remarks
Start permissive	0	1. No S/D active 2. Liquid in Scrubber 3. Inlet isolation valve open 4. Line A heater closed 5. Line B heater closed 6. System pressurized	NA	2	1. not UZS-4513.Y 2. not 45-LIC4826.WL 3. 45-HS4820.BCH 4. 45-XY4848.BCL 5. 45-XY4850.BCL 6. 45-PIC4831.YX> FUEL_GAS_SP	ELSE	1	Start order to be confirmed by operator. FUEL_GAS_SP can be adjusted by operator, default value 13,7 barg.
System not available	1	1. Warning on OS	1. 45-XA 9299	END	NA	NA	NA	
Set System in auto	2	1. Set system in auto	Set auto pin on 1. 45-XY4848 2. 45-XY4850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-LIC4827 8. 45-HS4822 9. 45-LIC4826 10. 45-HS4821 11. 45-LIC4828 12. 45-HS4823 11. 45-PIC4831 12. 45-HS4820 14. 45-TDIC4848 15. 45-TDIC4850 16. 45.XV4804 17. 45-XL4805	3	NA	NA	NA	
Check Line A spill-off	3		1. 45-HS4829.XL	4	1. 45-HS4829.BCL	NA		
Check Line B block valves	4	1. Open to Metering 2-4. Open to liquid outlet. 5-6. Check that level control is in auto	1. 45-HS4825.XH 2. 45-HS4823.XH 3. 45-HS4821.XH 4. 45-HS4822.XH 5. 45-XV4804.XH 6. 45-XV4805.XH	5	1. 45-HS4825.BCH 2. 45-HS4823.BCH 3. 45-HS4821.BCH 5. 45-LIC4826.BA 6. 45-LIC4828.BA 7. 45-XV4804.BCH 8. 45-XV4805.BCH	N/A	N/A	Normally this sequence is run after the stop or pressurization sequences. In that case these valves are already opened.
Check Line A block valves	5	1. Close to Metering	1. 45-HS4824.XL	6	1. 45-HS4824.BCL	NA	N/A	
Open Line B spill-off	6		1. 45-HS4830.XH	7	1. 45-HS4830.BCH			
Start Heater B	7		1. 45-XY4850.XP1H	8	1. 45-XY4850.BCH	NA	NA	
Wait for stabilization	8			9	1. 45-TDIC4850.YX=.YR			
Wait for stabilization	9			10	1. 45-TDIC4848.YX=.YR 2. timer=60s	NA	8	Timer value to be adjusted

Wait for Compressor purge and inlet valves to open	10	1. Indicate Fuel gas available to OS	1. 45-XA9101A.X	11	1. 27-HS4806.BCH 2. 27-ZSH4005.BCH		Override 45-HS4829.BCL to indicate that Line A is up and fuel gas available. The program should anticipate the inclusion of the following measures to avoid an unacceptable fluctuation in gas temperature when turbine starts. 1. Temporary change in PID parameters (45-TDIC4848 and 45-TDIC4850). 2. Temporary change in set points.
Close Line B spill-off	11	1. Close Line B spill-off line	1. 45-HS4830.XL	12	1. 45-HS4830.BCL	N/A	27-HS4006.BCH indicates flow either to flare or to turbine
	12		Release Auto pin on 1. 45-XY4848 2. 45-XY4850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-LIC4827 8. 45-HS4822 9. 45-LIC4826 10. 45-HS4821 11. 45-LIC4828 12. 45-HS4823 11. 45-PIC4831 12. 45-HS4820 13. 45-HS4806 14. 45-TDIC4848 15. 45-TDIC4850	END			
END	END	NA	NA	NA	NA	NA	NA

Sequence 3, Stop Fuel Gas System.

Phase	Step	Actions	Outputs to	Next step	Transition Condition	Jump condition	Jump to step	Remarks
Start permissive	0	*NOTE 1	NA	1	Operator confirmation		1	Start order to be confirmed by operator
Set System in auto	1	1. Set system in auto	1. 45-XY4848 2. 45-XY4850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-LIC4827 8. 45-HS4822 9. 45-LIC4826 10. 45-HS4821 11. 45-LIC4828 12. 45-HS4823 13. 45-PIC4831 14. 45-HS4820 15. 45-TIC4848 16. 45-TIC4850 17. 45-PIC4831	2	NA	NA	NA	
Open spill-off line	2		1. HS-XY4829.XH 1. HS-XY4830.XH	3	NA	NA		Operation of Spill-off valves is interlocked with heater outlet temperature. see page 45-101 and 45-102
Stop Heater A and B	3		1. 45-XY4848.XP1L 2. 45-XY4850.XP1L	4	1. 45-XY4848.BCL 2. 45-XY4850.BCL	NA	NA	
	4	Check opening of spill-off Line A.		5	1. 45-HS4829.BCH or 45-TI4848.BXL	NA		
	5	Check opening of spill-off Line B.		6	1. 45-HS4830.BCH or 45-TI4850.BXL	NA		
Close liquid outlet isolation valves	6		1. 45-HS4821.XL 2. 45-HS4822.XL 3. 45-HS4823.XL	7	1. 45-HS4821.BCL 2. 45-HS4822.BCL 3. 45-HS4823.BCL	NA		
Wait for temperature stabilization	7			8	1. 45-TI4848.BXL 2. 45-TI4850.BXL	NA		
Wait for temperature stabilization	8			9	1. 45-TI4848.BXL 2. 45-TI4850.BXL 3. timer=60s	ELSE	7	Timer value to be adjusted
Close spill-off valves	9		1. 45-HS4829.XL 2. 45-HS4830.XL	10	1. 45-HS4829.BCL 2. 45-HS4830.BCL	NA		

Reset Forced auto	10		Release auto pin on: 1. 45-XY4848 2. 45-XY4850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-LIC4827 8. 45-HS4822 9.45-LIC4826 10.45-HS4821 11.45-LIC4828 12.45-HS4823 13.45-PIC4831 14.45-HS4820 15.45-HS4806 16.45-TIC4848 17.45-PIC4831	END					
END	END	NA	NA	NA	NA	NA	NA	NA	

Note 1:
If turbine is running (signal from C09) and valve 45-XV4806A/ B
opened, then stop sequence will not start and alarm 45-XA9203 will
be activated.

Sequence 4, Change to Fuel Gas System Line A

Phase	Step	Actions	Outputs to	Next step	Transition Condition	Jump condition	Jump to step	Remarks
Start permissive	0	1 No S/D 2 Line B running 3 Line A not running	NA	2	1. not 79-UZS4513.Y 2. 45-XA9101B 3. not 45-XA9101A	ELSE	1	option to consider: Duty-Standby configuration of Line A and B with automatic change of line in case of equipment failure
System not available	1	1. Warning on OS	1. 45-XA 9299	END	NA	NA	NA	
Set system in auto	2	1. Set system in auto	Set auto pin on: 1. 45-XY4848 2. 45-XY850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-HS4822 7. 45-HS4823 8. 45-LIC4827 9. 45-LIC4828 10. 45-TDIC4848 11. 45-TDIC4850	3	NA	NA	NA	
Open Line A valves	3	1. Open Line A spill-off line. 2. Open to liquid outlet 3. Check that level control is in auto	1. 45-HS4829.XH 2. 45-HS4822.XH	4	1. 45-HS4829.BCH 2. 45-HS4822.BCH 3. 45-LIC4827.BA	NA	N/A	
Start heater Line A	4	1. Start heater Line A. 2. Lock External Set Point.	1. 45-XY4848.XP1H 2. 45-TIC4848.LX	5	1. 45-XY4848.BCH	N/A	N/A	Increased Set Point. XR= 45° C External SetPoint to be adjustable from OS
Wait for stabilization	5			6	1. 45-TDIC4848.YX=.YR	NA		Within 5%
Wait for stabilization	6			7	1. 45-TDIC4848.YX=.YR 2. timer=60s	ELSE	5	Timer value to be evaluated
Open Line A	7	1. Open Line A	1. 45-HS4824.XH	8	1. 45-HV4824.BCH	NA	N/A	The program should anticipate the inclusion of the following measures to avoid an unacceptable rise in gas temperature when the flow in line B drops. 1. Temporary change in PID parameters (TDIC4848 and TDIC4850). 2. Temporary increase in set points.
Close Line A spill-off	8	1. Close Line A spill-off line	1. 45-HS4829.XL	9	1. 45-HS4829.BCL	NA	NA	

Sequence 4, Change to Fuel Gas System Line A

Phase	Step	Actions	Outputs to	Next step	Transition Condition	Jump condition	Jump to step	Remarks
Both Lines active	9	Wait for heater respond to increased flow.		10	1. 45-TDIC4848.YX≥.YR 2. Timer=60s	NA		Opening speed for HV-4824 to be adjusted on the actuator. Timer value to be evaluated
Open Line B spill-off	10	1. Open spill-off Line B	1. 45-HS4830.XH	11	1. 45-HS4830.BCH	NA	NA	
Close Line B	11	1. Close Line B	1. 45-HS4825.XL	12	1. 45-HS4825.XL	NA	NA	Closing speed for HV-4825 to be adjusted on the actuator.
S/D heater line B	12	1. S/D Heater 2. Reset Line A to internal SP	1. 45-XY4850.XP1L 2. 45-TIC4848.LI	13	1. 45-TI4850.BCL	NA		
Heater B cooling.	13	Wait for cool down		14	1. 45-TI4850.BXL	NA		
Heater B cooling.	14	Wait for cool down		15	1. 45-TI4850.BXL 2. timer=60sek	ELSE	13	
Close spill-off Line B	15	close spill-off	1. 45-HS4830.XL	16	1.45-HS4830.BCL	NA		
Wait for stabilization	15			17	1. 45-TDIC4848.YX=.YR	NA		Within 5%
Wait for stabilization	17			18	1. 45-TDIC4848.YX=.YR 2. timer=60s	ELSE	15	Timer value to be evaluated
Reset auto	18	Reset Lock automatic	Release auto pin on: 1. 45-XY4848 2. 45-XY850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-HS4822 7. 45-HS4823 8. 45-LIC4827 9. 45-LIC4828 10. 45-TDIC4848 11. 45-TDIC4850	END				
END	END	NA	NA	NA	NA	NA	NA	

Sequence 5, Change to Fuel Gas System Line B.

Phase	Step	Actions	Outputs to	Next step	Transition Condition	Jump condition	Jump to step	Remarks
Start permissive	0	1 No S/D 2 Line A running 3 Line B not running	NA	2	1. not 79-UZS4513.Y 2. 45-XA9201 3. not 45-XA9201	ELSE	1	option to consider: Duty-Standby configuration of Line A and B with automatic change over in case of equipment failure
System not available	1	1. Warning on OS	1. 45-XA 9299	END	NA	NA	NA	
Set system in auto	2	1. Set system in auto	Set auto pin on: 1. 45-XY4848 2. 45-XY850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-HS4822 7. 45-HS4823 8. 45-LIC4827 9. 45-LIC4828 10. 45-TDIC4848 11. 45-TDIC4850	3	NA	NA	NA	
Open Line B valves	3	1. Open Line B spill-off line. 2. Open to liquid outlet 3. Check that level control is in auto	1. 45-HS4830.XH 2. 45-HS4823.XH	4	1. 45-HS4830.BCH 2. 45-HS48223.BCH 3. 45-LIC4828.BA	NA	N/A	
Start Line B	4	1. Start heater Line B. 2. Lock External Set Point.	1. 45-XY4850.XP1H 2. 45-TIC4850.LX	5	1. 45-XY4850.BCH	N/A	N/A	
Wait for stabilization	5			6	1. 45-TDIC4850.YX=.YR	NA		
Wait for stabilization	6			7	1. 45-TDIC4850.YX=.YR 2. timer=60s	ELSE	5	Timer value to be evaluated
Open Line B	7	1. Open Line B	1. 45-HS4825.XH	8	1. 45-HV4825.BCH	NA	N/A	The program should anticipate the inclusion of the following measures to avoid an unacceptable rise in gas temperature when the flow in line B drops. 1. Temporary change in PID parameters (TDIC4848 and TDIC4850). 2. Temporary increase in set points.
Close Line B spill-off	8	1. Close Line B spill-off line	1. 45-HS4830.XL	9	1. 45-HS4830.BCL	NA	NA	

Both Lines active	9	Wait for heater respond to increased flow.		10	1. 45-TID4850.YX≥.YR 2. Timer=60s	NA		Opening speed for HV-4825 to be adjusted on the actuator. Timer value to be evaluated
Open Line A spill-off	10	1. Open spill-off Line A	1. 45-HS4829.XH	11	1. 45-HS4829.BCH	NA	NA	
Close Line A	11	1. Close Line A	1. 45-HS4824.XL	12	1. 45-HS4824.BCL	NA	NA	Closing speed for HV-4824 to be adjusted on the actuator.
S/D heater line A	12	1. S/D Heater 2. Reset Line B to internal SP	1. 45-XY4848.XP1L 2. 45-TIC4850.LI	13	1. 45-XY4848.BCL	NA		
Heater A cooling.	13	Wait for cool down		14	1. 45-TI4848.BXL	NA	NA	
Heater A cooling.	14	Wait for cool down		15	1. 45-TI4848.BXL 2. timer=120sek	ELSE	13	Timer value to be evaluated
Close spill-off Line A	15	1. Close spill-off	1. 45-HS4829.XL	16	1. 45-HS4829.BCL	NA		
Wait for stabilization	15			17	1. 45-TDIC4850.YX=.YR	NA		Within 5%
Wait for stabilization	17			18	1. 45-TDIC4850.YX=.YR 2. timer=60s	ELSE	15	Timer value to be evaluated
Reset auto	18	Reset Lock automatic	Release auto pin on: 1. 45-XY4848 2. 45-XY850 3. 45-HS4830 4. 45-HS4825 5. 45-HS4829 6. 45-HS4824 7. 45-HS4822 7. 45-HS4823 8. 45-LIC4827 9. 45-LIC4828 10. 45-TDIC4848 11. 45-TDIC4850	END				
END	END	NA	NA	NA	NA	NA	NA	

Sequence 6, Pressurization Fuel Gas System.

Phase	Step	Actions	Outputs to	Next step	Transition Condition	Jump condition	Jump to step	Remarks
Start permissive	0	1. No S/D active 2. Liquid in Scrubber 3. Inlet isolation valve closed 4. Outlet isolation valve closed	NA	2	1. not UZS-4513.Y 2. not 45-LIC4826.WL 3. 45-HS4820.BCL 4. 45-HS4806.BCL	ELSE	1	Start order to be confirmed by operator. The cold system will generate alarms for abnormal system conditions (E.g. PALL4859). These alarms shall be suppressed with standard procedure.
System not available	1	1. Warning on OS	1. 45-XA 9299	END	NA	NA	NA	
Set System in auto	2	1. Set system in auto	set auto pin high on: 1. 45-HS4830 2. 45-HS4825 3. 45-HS4829 4. 45-HS4824 5. 45-LIC4827 6. 45-HS4822 7. 45-LIC4826 8. 45-HS4821 9. 45-LIC4828 10. 45-HS4823 11. 45-PIC4831 12. 45-HS4820	3	NA	NA	NA	
Isolate System	3	1. Force 45-PV4831 to Closed 2-3. Close spill of lines	1. 45-PIC4831.FSL 2. 45-HS4829.XL 3. 45-HS4830.XL	4	1. 45-PIC4831.BS 2. 45-HS4829.BCL 3. 45-HS4830.BCL	NA		
Open internal block valves	4	1-2. Open to Metering 3-5. Open to liquid outlet. 6-7. Check that level controls are in auto	1. 45-HS4824.XH 2. 45-HS4825.XH 3. 45-HS4821.XH 4. 45-HS4822.XH 5. 45-HS4823.XH	5	1. 45-HS4824.BCH 2. 45-HS4822.BCH 3. 45-HS4821.BCH 5. 45-LIC4826.BA 6. 45-LIC4827.BA 7. 45-LIC4828.BA	N/A	N/A	
Open inlet isolation valve	5	1. Open 45-HS4820	1. 45-HS4820.XH		1. 45-HS4820.BCH	NA		
Pressurize	5	1. Release force to close 2. Set Pressure control valve in auto mode	1. 45-PIC4831.FSL=0 2. 45-PIC4831.LA	6	1. 45-PIC4831.BA			Ramp function to be considered if I-function in 45-PIC4831 does not give a smooth opening of the valve. Note: 1. 45-PIC4831.X is located after metering. 2. After pressurization 45-PV4831 will close.

Wait for pressure to build up	6			7	1. 45-PIC4831.YX= FUEL_GAS_SP	NA		Process=Set point. FUEL_GAS_SP can be adjusted by operator, default value 13,7 barg.
Wait for stabilization	7				8 1. 45-PIC4831.YX> FUEAL_GAS_SP 2. timer=10s	ELSE	6	Timer value to be adjusted. As long as outlet valves remain closed measured value will overshoot set point. FUEL_GAS_SP can be adjusted by operator, default value 13,7 barg.
Reset Forced auto	8		Release Auto pin on 1. 45-HS4830 2. 45-HS4825 3. 45-HS4829 4. 45-HS4824 5. 45-LIC4827 6. 45-HS4822 7. 45-LIC4826 8.45-HS4821 9.45-LIC4828 10.45-HS4823 11.45-PIC4831 12.45-HS4820	END		NA		
END	END	NA	NA	NA	NA	NA	NA	